VIGNESH R

Aspiring Embedded Software Engineer
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SUMMARY

Electrical and Electronics Engineering graduate with a strong interest in Embedded Software Development. Skilled in Embedded C, C++, STM32, Arduino and microcontroller-based application development. Familiar with peripheral interfacing, embedded firmware development and IoT-based projects. Passionate about learning embedded technologies and seeking an opportunity to contribute as an Embedded Software Engineer while continuously improving technical skills.

TECHNICAL SKILLS

- Programming Languages: C, Embedded C, C++, Java
- Microcontrollers: Arduino, STM32
- Tools & IDEs: Arduino IDE, Keil uVision, Proteus (simulation), MI Power
- Protocols & Communication: I2C, GSM/SMS-based alerting
- Core Concepts: Embedded Systems, Version Control (Git)

PROJECTS

Gas Detection System — *Arduino, GSM, MQ Sensors*

- Built a real-time gas detection and alert system using Arduino and a GSM module, using MQ-series sensors to detect LPG, CO and methane.
- Implemented SMS alerting logic to instantly notify users when gas concentration crossed safe thresholds.
- Managed source code and collaborated with a team using Git.

Home Automation System — *STM32*

- Developed firmware on STM32 to control and monitor home appliances based on live sensor input.
- Integrated motion sensors for occupancy/object detection and LM35 temperature sensors for environment monitoring.
- Wrote automatic control logic to improve home safety and energy efficiency.

Digital Dice — *Embedded C*

- Designed a microcontroller-based digital dice generating random values via embedded C logic.
- Drove a 7-segment display to show dice values 1–6, triggered by push-button input.

INTERNSHIP

United IT and CADD Division, Salem Sept 2025

- Worked on IoT and microcontroller-based embedded applications during internship.
- Interfaced sensors and actuators with embedded hardware for monitoring applications.
- Assisted in implementing GSM-based SMS alert functionality.
- Gained practical exposure to embedded programming, circuit interfacing and debugging.

EDUCATION

B.E. — EEE - KSR Institute for Engineering and Technology, Namakkal 2022 – 2026
CGPA: 7.5

HSC — Kongu Higher Secondary School, Karur 2021 – 2022

SSLC — Kongu Higher Secondary School, Karur 2019 – 2020

CERTIFICATIONS

- Value Added Certification in Arduino Programming using Arduino IDE — San Tech Innovation
- Organizational Behaviour (72%) — NPTEL

EXTRACURRICULAR ACTIVITIES

- Participated in Smart India Hackathon (SIH) 2024, showcasing problem-solving and teamwork skills.
- Presented a paper at a Paper Presentation event, Government Engineering College, Erode.